

Part I: Short Framing on Climate Change

Please read the following text carefully. The technology explained on the next page refers to it:

According to the latest report by the Intergovernmental Panel on Climate Change (IPCC), the global average surface temperature has risen at a rate of 1°C (1.8°F) compared to 1900. It is extremely likely that this effect is caused by increased emissions of greenhouse gases such as carbon dioxide (CO₂). Greenhouse gas emissions are released for example through the burning of coal, oil and gas. One degree may not seem like much, but it already has an impact, which can worsen in the future as the temperature continues to rise:

- Increasingly extreme weather events, like heavy rainfall and storms.
- The oceans are warming due to human influence, which is leading to the melting of the polar ice caps.
- Increased acidity of the ocean, because the water takes up too much of the released CO₂, which is leading to the destruction of corals reefs worldwide.

If greenhouse gas emissions continue as they are right now, by 2100 the global temperature could rise up to 3.3 – 5.7°C (5.94 – 10.26°F) compared to before 1900.

Part II: non-fictional technology: 'Stratospheric Aerosol Injection'

Please read the following text carefully. The questions later in the study refer to it:

One possibility to counteract global warming is to reflect incoming heat from the sun brought by sunlight back into space to prevent further warming of land and oceans. This is called Solar Radiation Management. One specific technique would be "Stratospheric Aerosol Injection" using sulfur particles that are being released in higher regions of the atmosphere, which reflect sunlight partially back out into space before it warms the earth. Sulfur is a chemical element found in almost all life forms. It, for example, gives eggs their distinctive smell. When a volcano erupts, sulfur particles are also released into the atmosphere, where they produce a cooling effect, which was demonstrated by past volcanic eruptions. This could be mimicked by directly releasing sulfur particles into the air, for example by airplane, about 15 kilometers (50.000 feet) from the Earth's surface. There the particles would disperse and reflect partially the incoming sunlight. The benefits of "Stratospheric Aerosol Injection" (SAI) would be:

- feasible and potentially very effective technique
- compared to other technologies the costs of SAI are low
- SAI could be deployed rather soon and would start to reduce temperatures within one year

However, this technique also comes with certain risks and uncertainties:

- once started SAI needs to be deployed for decades or centuries – otherwise temperature could rise even more dramatically within just a few years
- SAI does nothing to counter the inherent cause of climate change – the concentration of carbon dioxide (CO₂) in the atmosphere - and therefore it does not reduce the acidity of the ocean
- political and social conflicts over its use, as one country could deploy SAI, but affect the whole Earth

As the potential side effects have not yet been fully resolved, this technique would only be an emergency solution in the case that the temperature actually rises close to the 2°C (3.6°F) target, which nearly all countries in the world have agreed upon. In order to achieve the target, the deployment of "Stratospheric Aerosol Injection" would have to start within the next few years. It does not change the fact, that we have to change our lifestyles and transform the economy in a way that limits global warming to 2°C (3.6°F) or lower until 2100.